Université Libanaise

Faculté des Sciences Économiques et de Gestion Section III

Département :	Informatique de Gestion	**Année Univ. :**	2024–2025
Matière :	User Interface	**Niveau / Sem. :**	M2 / Semestre S2
Responsable :	Dr. Adolf Abdallah	**Durée :**	90 minutes
Date d'examen :	16/06/2025	**Documents :**	Non Autorisés

OFFICIAL EXAM SOLUTION KEY (PROBLEM 1: OPEN QUESTIONS)

Question 1

5 Points

Define an XML Schema and briefly explain how it differs from a DTD.

OFFICIAL ANSWER:

XML Schema Definition: An XML Schema defines the structure, elements, and valid data rules of an XML document using standard XML syntax.

Key Differences from DTD:

- **Syntax:** XML Schema is written in valid XML syntax, whereas DTD uses its own unique, non-XML text syntax.
- **Data Types:** XML Schema supports rich data types (integers, dates, booleans), while DTD only supports basic text strings (`#PCDATA`).
- **Namespaces:** XML Schema fully supports XML Namespaces to avoid element conflicts; DTD does not support them natively.

Question 2

5 Points

List three advantages of using XML in web development.

OFFICIAL ANSWER:

- **Separates data from presentation:** Data is safely stored and decoupled from any specific visual styling or display logic.
- **Simplifies data sharing:** Its software/hardware-independent plain text format allows totally incompatible systems to exchange data smoothly.
- **Extensible framework:** It features no predefined tags; developers can freely create custom tags tailored precisely to their data structure.

Question 3

10 Points

Given the specified DTD rules, write a well-formed XML document that follows them.

OFFICIAL ANSWER:

```
<?xml version="1.0" encoding="UTF-8"?>
<COURSELIST>
    <COURSE CREDITS="3">
        <NAME>User Interface</NAME>
        <CODE>INF301</CODE>
        <INSTRUCTOR>Dr. Adolf Abdallah</INSTRUCTOR>
    </COURSE>
</COURSELIST>
```

Question 4

10 Points

Describe three key differences between HTML and XML.

OFFICIAL ANSWER:

- **Purpose:** HTML is strictly designed to **display data** (focusing on visual presentation), whereas XML is designed to **store and transport data** (focusing on structural content).
- **Tags:** HTML provides standard, predefined structural tags (e.g., `<html>`, `<h1>`); XML contains no predefined tags and relies entirely on custom, user-defined elements.
- **Syntax Strictness:** HTML is highly forgiving with missing closing tags and case sensitivity; XML enforces a strict syntax requiring full case sensitivity and exact matching closing tags.

Question 5

10 Points

Explain the concept of "Usability" in UI design and name its three core components.

OFFICIAL ANSWER:

Definition: Usability is a vital quality attribute that evaluates how easy, efficient, and natural a digital interface is for human users to accomplish their exact target goals.

Three Core Components:

- **Effectiveness:** The accuracy and completeness with which users successfully fulfill their intended tasks.
- **Efficiency:** The speed, effort, and resource optimization required by the user to execute those tasks.
- **Satisfaction:** The personal comfort, aesthetic acceptability, and general positive attitude of users interacting with the interface.

Question 6 *15 Points*

Discuss Norman's Seven Principles of UI Design with brief examples for any two of them.

OFFICIAL ANSWER:

The 7 Principles: (1) Discoverability, (2) Feedback, (3) Conceptual Model, (4) Affordances, (5) Signifiers, (6) Mappings, and (7) Constraints.

Detailed Core Examples:

- **Feedback:** A visible loading spinner or progress bar displaying instantly upon hitting a "Submit" button ensures the user immediately knows the system is processing their action.
- **Constraints:** Disabling or graying out a form's "Next" button until all mandatory text inputs are accurately completed restricts the user from accidentally executing an incomplete, error-prone submission.

Question 7 *15 Points*

In XPath, explain the meaning and output of each of the given expressions.

OFFICIAL ANSWER:

Expression	Meaning	Output
`//course/name`	Finds all <name> elements that are direct children of a <course> parent, anywhere in the XML document tree.	A collection/list of the matching <name> element nodes.
`/library/book[2]/title`	Starts at the root node <library>, targets its absolute 2nd <book> child element, and navigates down to its structural <title>.	The individual <title> node of the second book.
`//book[@lang='en']`	Scans the whole document relatively to match all <book> tags possessing an internal attribute called 'lang' with a value of 'en'.	A collection of the filtered <book> element nodes.
`//student[score>90]`	Selects all <student> elements anywhere in the tree containing a direct child node named <score> whose numeric value is strictly above 90.	A collection of the matching <student> element nodes.
`/school/class[1]/student[1]/@id`	From root <school>, traverses down to the first <class>, accesses the first <student> inside, and points to their 'id' attribute tag.	The exact text value inside that specific student's 'id' attribute.

Question 8

10 Points

What is "Direct Manipulation" in UI, and what are its two main components?

OFFICIAL ANSWER:

Definition: Direct Manipulation is an interactive design paradigm where users execute actions on digital objects using tactile, physical, real-world metaphors (e.g., dragging items, dragging to trash, pinching to zoom) instead of typing textual terminal commands.

Two Main Components:

- **Continuous representation of objects:** Crucial interactive elements remain permanently visible on screen.
- **Rapid, incremental, reversible actions:** Physical interface interactions yield instant visual feedback and are effortless to undo.

Question 9

10 Points

What is a "Heuristic Evaluation"? Explain its purpose in interface design.

OFFICIAL ANSWER:

- **Definition:** A heuristic evaluation is a structured inspection method where usability and UX experts systematically evaluate an interface design against a solid framework of established, recognized principles ("heuristics").
- **Purpose:** To quickly and cost-effectively identify critical usability flaws and layout conflicts early in the product lifecycle, enabling rapid iterative fixes before writing production code.

Question 10

10 Points

List and briefly explain any four of Nielsen's Ten Usability Heuristics.

OFFICIAL ANSWER:

- **1. Visibility of system status:** Ensure users are kept informed of system operations in real time via clear, immediate status updates.
- **2. Match between system and real world:** Implement clear, natural everyday terminology and design metaphors familiar to users, bypassing code jargon.
- **3. User control and freedom:** Integrate explicitly marked "emergency exits" (such as Undo, Redo, and Cancel buttons) to help users easily backtrack from accidental errors.
- **4. Consistency and standards:** Maintain cross-platform uniformity in text labels, interaction behavior, and icon glyphs so users avoid cognitive confusion.